# 1 Complex Vector Encodings and Convolution

An alternative approach is to utilize two distinct vectors ($A$ and $B$) to store the coefficients of complex numbers. These vectors can be multiplied using convolution:

$$A = [a_1, a_2, \ldots, a_n]^T$$
$$B = [b_1, b_2, \ldots, b_n]^T$$

These vectors represent $n$ complex numbers or their respective coefficients.

## 1.1 Security Assertions

This construction appears robust based on the following assertions:

- It is computationally infeasible to directly calculate the coefficients $a_1, \ldots, a_n$, as no initial assumptions can be made regarding their values, and the Discrete Fourier Transform (DFT) remains secure against direct inversion attacks.
- While the system can theoretically be inverted (yielding a zero-test), the encoding prevents any structural leakage. Because the operations are strictly over integers, the $z/z'$ encoding is not compromised. The only scenario where the system of linear equations (SLE) becomes unsolvable is when encountering a zero, but since this zero does not originate from $a^2 + b^2$, the $z/z'$ encoding remains secure.
- A potential vulnerability lies in the convolution $Z = A * A + B * B$. If the plaintext $m = 0$, $Z$ evaluates to the null vector ($0'$). It remains to be formally verified whether this state leaks the condition $a^2 + b^2 = 0$.

# 2 Symmetry and Basis Vector Analysis

Empirical tests over a prime modulus ($p = 25261$ with parameters $a = 12631, b = 1543, u = 16700$) reveal recurring identical values within the resulting vectors. However, unlike the previously analyzed $C^2$ encoding where specific variable values were exposed, the exact values here remain entirely unknown. While symmetries (identical values) are observable, the underlying scalars are concealed.

A critical issue arises regarding the basis vectors $b_2$ and $b_3$. The vectors exhibit specific symmetries:

$$A = [a, x, y], \quad B = [b, j, k]$$
$$A = [a, y, x], \quad B = [-b, -k, -j]$$

This symmetry potentially allows an adversary to isolate the random scalars $r_1$ and $r_2$. The system yields known linear combinations:

$$x \cdot (r_1^n + r_2^n) \tag{1}$$
$$y \cdot (r_1^n - r_2^n) \tag{2}$$

It is hypothesized that knowing $(r_1^n + r_2^n)$ does not directly compromise the system, as it is mathematically distinct from $(r_1 + r_2)^n$.

## 3 Formal Verification via SMT Solvers

Given the intractability of manually proving the security of these complex DFT algorithms, automated formal verification is required. Since an infinite number of algebraic equations can be generated, a tool like Microsoft's Z3 SMT-solver could be utilized.

By defining the constraints (e.g., multiplication rules where $\text{mul}(b_i, b_j) = 0$ for $i \neq j$) and supplying the solver with standard addition and scalar multiplication rules, one could test if the solver can successfully extract the unknowns $r_1$ and $r_2$.

## 4 Impossibility Proof of Parameter Extraction

**Theorem (Conjecture):** It is impossible to extract the random scalars $r_1$ and $r_2$.

*Proof Outline:* Assume the variables $a, b, x, y, j, k$ are random and independent. If an adversary could successfully extract $r_1$ and $r_2$, they would simultaneously break the underlying $z/z'$ encoding. By identifying $r_1$ and $r_2$, the adversary could set $r_1 = 0$, forcing a zero-state ($c \cdot c - r_1 \cdot c = 0$). This would permit the adversary to compute the square root of $-1$, which is mathematically presumed to be hard/impossible without the prime factorization. Therefore, the extraction of $r_1$ and $r_2$ must be impossible.

## 5 The Ratio Attack on Zero-Plaintexts

Another theoretical attack vector emerges when $m = 0$. In this state, vector $B$ can be divided by vector $A$, yielding a ratio vector $K$ ($A = K \cdot B$). The ratio $K$ reflects the relationship between the real and imaginary parts, which remains constant when $|z| = 0$.

This vector $K$ takes the form $[0, t, -t]^T$, where $\left(\frac{p+1}{2}\right) \cdot t = b$, resulting in the zero-condition $\left(\frac{p+1}{2}\right)^2 + t^2 = 0$. While this exposes the structural zero, the actual value of $t$ cannot be efficiently extracted from $K$. The adversary can only extract $t^2$ (via $K \cdot K$), but cannot compute $t$ itself without the ability to compute modular square roots. Thus, the system arguably remains secure.